

Continuous Noninvasive Wearable Multi-spectral Optical Sensor for Multi-Biomarker Research

Patricia D. Flores[^], Haoran Zhang[^], Emily E. Grammer[†], Hailey R. Russo[^], Macy E. Stahl[†], Robert H. Thiele[^], Kyle C. Quinn[^], Jason D. Allen[†], Arthur Weltman[†], Afsaneh Doryab^{*}, Sibylle Kranz[†], Amanda Watson[^]

[^]Dept. of Electrical and Computer Engineering, University of Virginia

^{*}Dept. of Systems and Information Engineering, University of Virginia

[^]Dept. of Anesthesiology, University of Virginia, [†]Dept. of Kinesiology, University of Virginia

{sjsy7da, hz7fs, erk9nk, xsu9rn, mes4hm, rht7w, bak6sp, ja6af, alw2v, ad4ks, sk4gd, aawatson}@virginia.edu

Abstract—Noninvasive methods for continuous wearable biomarker detection research in the infrared spectrum remain conceptual due to the size and power consumption of infrared sensors. This paper introduces IR2, a continuous noninvasive wearable device that uses multi-spectral optical sensing to enable multi-biomarker research. We characterized IR2 through a series of benchtop evaluations assessing device variability, environmental robustness, and power performance. Furthermore, we deployed IR2 in four laboratory and clinical environments spanning exercise, clinical, nutrition, and eating applications, collecting more than 560 hours of optical data from 79 participants. These deployments identified and informed the resolution of several design limitations, including BLE dropout, insufficient transimpedance amplifier gain, inadequate battery life, and enclosure vulnerabilities, providing practical insights for the development of future wearable spectroscopy systems.

Index Terms—noninvasive, wearable, biomarker, optical sensor

I. INTRODUCTION

Wearable biosensors have gained popularity due to their noninvasive and continuous monitoring capabilities [1], with optical sensors operating in the visible to near-infrared (NIR) spectrum being the most widely used [2], [3]. However, these sensors cannot detect all biomarkers, as analytes such as lactate and proteins exhibit absorption features in the NIR/short-wave infrared (SWIR) range [4], [5]. Although SWIR-based biosensors show potential for expanding wearable biomarker detection, wearable implementations remain largely conceptual because the large footprint and high power requirements of SWIR emitters and detectors make it difficult to design a device spanning the visible to SWIR spectrum [6]. Thus, existing wearable optical devices are limited to visible-only [7]–[11], NIR/SWIR-only [6], or narrow-wavelength sensing [12].

In this paper, we present IR2, a continuous noninvasive wearable multi-spectral optical sensor for multi-biomarker research. IR2 is a wrist-worn device that noninvasively probes the epidermis to support diverse biomarker sensing applications, using light spanning the visible to SWIR wavelength range (400–1750nm). Our contributions are summarized as follows: (1) A wrist worn visible to SWIR multi-spectral optical sensor, (2) Experiment characterization under four testbenches to verify device performance and simulate various environments (3) Deployment across four human studies

totaling 79 participants and more than 560 hours of use (4) Practical lessons learned to improve device performance.

II. DEVICE HARDWARE

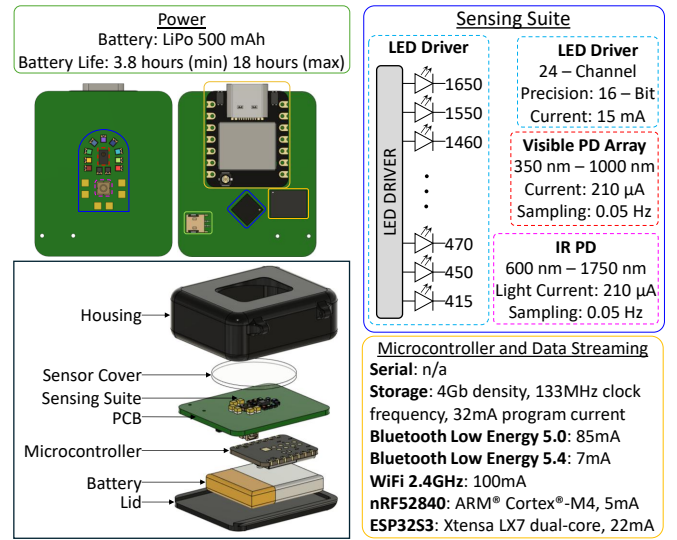


Fig. 1. Device Design

This section covers IR2’s sub-systems: sensing, data streaming, battery, and housing. Fig. 1 shows the full device design.

A. Sensing Suite

The sensing suite of IR2 features 17 light-emitting diodes (LEDs), nine in the visible light spectrum and eight in the NIR/SWIR light spectrum, a visible photodiode array, and a broadband Indium Gallium Arsenide (InGaAs) photodiode. Nine visible LEDs span 415–660nm (~30nm spacing) and eight NIR/SWIR LEDs cover the infrared (IR) range (~100nm spacing) for maximum spectral coverage. Compared to the wearable spectrometer used for skin tone [9], nutrition [11], and blood pressure [10] sensing, IR2 is 5% smaller, retains the same visible sensor, and adds NIR/SWIR sensing.

1) **LEDs:** IR2 features 17 surface-mount device (SMD) LEDs controlled by an LED driver to adjust brightness, with center wavelengths of 415, 450, 470, 500, 530, 567, 599, 633, 660, 850, 940, 1050, 1200, 1300, 1460, 1550, and 1650nm.

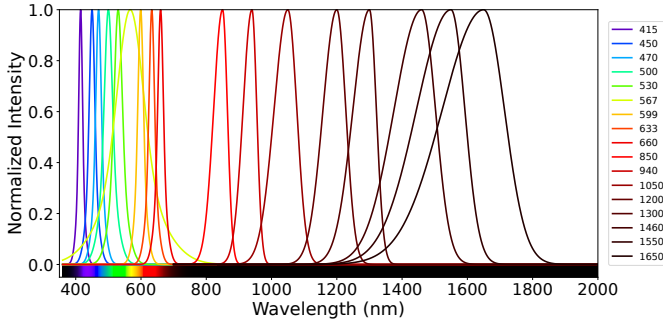


Fig. 2. LED Response.

Fig. 2 shows the full LED responsivity. The 415–940nm LEDs are LUXEON Z-Colorline LEDs and the 1050–1650nm LEDs are Marktech SMD SWIR LEDs, selected for their commercial availability, compact footprint ($1.3 \times 1.7\text{mm}$ and $1.6 \times 1.6\text{mm}$ respectively), and broad spectral coverage. The 415–1650nm range was chosen to ensure photobiological safety, as wavelengths below 400nm risk UV exposure [13], capture analyte absorption in the IR range [4], [5], and align with the longest-wavelength SMD LED commercially available.

All 17 LEDs are controlled by the TLC5947 LED driver, which supports up to 24 LEDs at a max current of 15mA set by a current-limiting resistor. We selected the TLC5947 for its low power consumption, PWM dimming, SPI communication, and compact footprint ($4.8\text{mm} \times 4.8\text{mm}$).

2) *Detectors*: IR2 features two detectors because silicon photodiodes exhibit limited sensitivity beyond 1100nm, whereas InGaAs photodiodes maintain strong responsivity in the NIR/SWIR region. Combining the AS7341 spectral sensor, a silicon based sensor, with an InGaAs photodiode, IR2 achieves broad spectral coverage spanning the visible, NIR, and SWIR wavelengths (Fig. 3).

The AS7341 is an 11-channel I2C spectral color sensor covering a 350–1000nm wavelength range, featuring eight visible photodetectors, one NIR photodetector, and two additional channels: clear and flicker. The center wavelengths for the nine spectral channels are 415, 445, 480, 515, 555, 590, 630, 680, and 910nm (see Fig. 3). The clear channel provides broadband measurements for normalization and calibration, while the flicker channel detects periodic light variations.

The InGaAs P-type Intrinsic N-type (PIN) photodiode is the Marktech MTSM1346SMF1-100, a broadband photodiode covering the 600–1750nm spectral range. This photodiode has an overall size of $3\text{mm} \times 3\text{mm}$, an active area of 1mm^2 , and a peak responsivity of 0.74 occurring at 1650nm. Because the light current is in the microamp range, the photodiode output feeds a transimpedance amplifier with a gain of $1\text{M}\Omega$ and a bandwidth of 58kHz, communicating with the microcontroller via a 12-bit analog to digital converter (ADC).

3) *Footprint*: The sensing suite is centered on a printed circuit board (PCB) to ensure symmetric placement and to simplify the design of the housing for the device. Since the active area of the AS7341 is not centered on the component itself, the LEDs were strategically positioned to account for

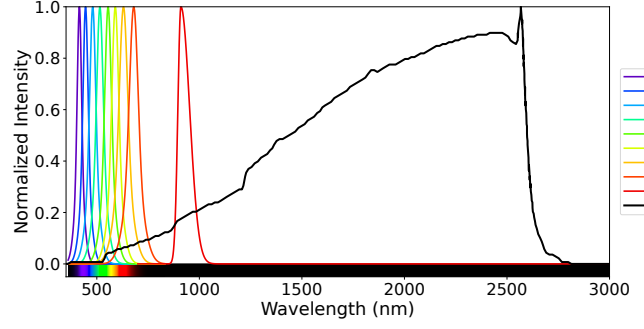


Fig. 3. PD Responsivity.

this offset. Specifically, LEDs with shorter wavelengths were positioned closer to the active area of the AS7341, as optical penetration depth generally increases with wavelength in the 400–1000nm range [14]. Decreasing the source-detector separation for shorter wavelengths helps to improve the collection of reflected light and maintain adequate signal strength.

B. Data Streaming & Microcontroller

Different deployment settings have distinct constraints on mobility, power consumption, data reliability, and study duration, which require different data streaming methods. IR2 supports five data streaming methods: Serial, NAND Storage, Bluetooth Low Energy (BLE) 5.0, BLE 5.4, and 2.4GHz WiFi. These methods fall into three categories: wired (Serial), wireless without real-time data verification (NAND storage), and wireless with real-time data verification (BLE and WiFi). Serial and NAND storage data streaming systems are available on both models of the device. However, BLE 5.0 and 2.4GHz WiFi require the Seeeduino XIAO ESP32S3 and BLE 5.4 requires the Seeeduino XIAO nRF52840. We chose to develop two models, since each microcontroller provides different benefits. For example, a device with the ESP32S3 offers added WiFi data streaming, faster processing, and multi-cores, but at the cost of higher power consumption versus an nRF52840.

The MT29F4G01ABAFDWB is the NAND storage device on IR2. This storage device uses SPI communication, can hold up to 4Gb of data, features a 133MHz max clock frequency, allows for 100,000 PROGRAM/ERASE cycles, collects 11 days of spectral data, and can run between 2.7V and 3.6V. The low operating voltage and the large number of PROGRAM/ERASE cycles are important features, since IR2 runs on 3.3V with battery power and runs for multiple trials.

C. Battery

The device uses a 3.7V 500mAh Lithium-ion Polymer battery ($20\text{mm} \times 37\text{mm} \times 9.5\text{mm}$), compact enough to keep the device size small while providing up to 20 hours of operation.

D. Housing

The housing for IR2 is a $34.5\text{mm} \times 40.5\text{mm} \times 18.11\text{mm}$ FDM (Fused Deposition Modeling) printed PLA (Polylactic Acid) case designed to be comfortably worn on the wrist, protecting both the user and the device. The housing features a friction fit sliding lid for easy access and construction,

a 1.1mm thick, 25mm diameter GorillaGlass® window as a protective barrier between the device and the user, strap holders angled down by 100° for comfort and an ergonomic fit, and a conveniently accessible charging port. A temporary prototyping spacer sits between the lid and the device to ensure it is compressed all the way down, preventing shifting without compromising the readings from the sensing suite.

III. EVALUATION

a) Device-to-Device Comparison: This test evaluates the variation between six IR2 devices configured with identical LED and photodetector settings. We quantified inter-device variability as the group mean across all device combinations with a $\pm 10\%$ acceptable threshold since commercially available components are not high precision instruments. The coefficient of variance (CV) for the visible channels was 7.4%, which was lower than our acceptable threshold. However, the IR channel's 13% CV exceeded our threshold, likely due to the IR photodiode's lack of optical filters, increasing sensitivity to ambient noise and warranting further investigation.

b) Battery Experiment: This test experimentally characterizes the battery life of IR2 for each operational configuration. To minimize confounding factors, each test used identical LED and photodetector settings. Results are summarized in Tab. I, informing the device configuration for each deployment environment. The ESP32S3 with NAND storage configuration was excluded because the device has not been manufactured, and Serial was omitted as the wired connection powers IR2.

TABLE I
BATTERY LIFE AND CONNECTION TYPE BY DATA STREAMING PROTOCOL

Data Transfer Type	Microcontroller	Life Span	
		Theoretical	Experimental
NAND Storage	ESP32S3	~13 hrs	n/a
	nRF52840	~20 hrs	18 hrs
Bluetooth	ESP32S3	~4.3 hrs	4.38 hrs
	nRF52840	~14 hrs	13 hrs
WiFi	ESP32S3	~4 hrs	3.8 hrs

c) Temperature Experiment: The temperature experiment evaluates device performance across three distinct thermal environments: cold (1°C), room temperature (25°C), and hot (46.5°C). These conditions represent the thermal boundaries and nominal operating conditions of real-world wearable use. Readings were 1.9% and 1.3% higher in hot and cold environments, respectively, with no drift or recalibration required.

d) Fluid Study: This study examines the effect of surface moisture on device readings by comparing measurements from a dry device against one with water applied to the glass window, simulating perspiration encountered during daily wear and exercise. Spectral readings were 4.8% lower in low sweat conditions (0.05mL) and 8.9% lower in high sweat conditions (0.15mL) compared to dry conditions.

IV. REAL WORLD DEPLOYMENT

Following laboratory characterization, IR2 was deployed across four IRB approved studies representing diverse op-

erating environments and sensing requirements (Table II). These deployments informed multiple hardware iterations and provided insight into communication reliability, power management, housing design, and analog signal conditioning.

TABLE II
SUMMARY OF DATA COLLECTION STUDIES

Study Type	# of Participants	# of Hours Collected	Configuration
Lactate-Exercise	20	7.75	nRF52840 w/ BLE 5.4
Lactate-CICU	20	321	nRF52840 w/ Storage
Eating Detection	20	10	nRF52840 w/ Serial
Nutrition	19	228	nRF52840 w/ BLE 5.4

A. Exercise-Induced Lactate Monitoring

The goal of this study was to investigate lactate-associated optical changes under controlled conditions. Exercise was used to induce substantial lactate fluctuations, providing a reproducible means of collecting optical data during periods of meaningful physiological change, while IR2 noninvasively and continuously collected spectral data on 20 participants at the UVA Kinesiology lab. Tab. II shows the device configuration. We implemented the transimpedance amplifier system and decided to use Bluetooth 5.4 data streaming since participants were actively exercising for less than an hour, mitigating battery life concerns.

This deployment demonstrated that the housing maintained reliable operation despite exposure to perspiration during exercise. However, the BLE 5.4 connection occasionally dropped, resulting in data loss and motivating the future integration of onboard storage. Additionally, the original 60k Ω transimpedance amplifier gain proved insufficient, as the collected readings from human subjects only utilized 50% of the ADC's 12-bit range despite driving the IR LEDs at maximum brightness. We resolved this by increasing the gain to 1M Ω , which produced readings covering the range. These findings illustrate how deployment in active settings can reveal limitations in communication reliability and analog signal conditioning that may not be apparent during benchtop testing.

B. Lactate-CICU

The goal of this study, conducted at UVA Health was to evaluate lactate-associated optical changes during prolonged inpatient monitoring. The IR2 configuration can be seen in Tab. II. This configuration maximized power efficiency without external communication. Because data collection occurred overnight without opportunities for device replacement or recharging, the system required a wireless configuration capable of operating continuously for more than 18 hours to prevent data loss. To reduce the risk of accidental manipulation, the housing lid was temporarily secured using a biocompatible UV-cured adhesive, highlighting how clinical deployments prioritize uninterrupted operation over device compactness.

The original 290mAh battery achieved only 12 hours of battery life with the nRF52840 microcontroller and NAND

storage data streaming, six hours short of our 18-hour minimum. Since we were already running the lowest power consumption configuration, software optimizations could only give us a few extra minutes, not the several hours needed. This left increasing battery capacity as our only viable option. However, a compact, commercially available battery of at least 400mAh, at the same size as the 290mAh battery, proved difficult to source. We ultimately settled on a 500mAh battery, which was slightly thicker than the original.

C. Eating Detection

The goal of this study was to investigate whether wearable spectroscopy could capture optical changes associated with different levels of satiety during food consumption. We collected data on 20 participants using IR2, and its configuration can be seen in Tab. II. Serial communication provided a simple and reliable method for data collection despite requiring a wired connection since this was a seated study lasting less than 30 minutes. This deployment demonstrated that wired communication remains an effective and reliable solution for short-duration studies conducted in controlled environments.

D. Nutrition

This study investigated whether wearable spectroscopy could capture optical changes associated with postprandial nutrient dynamics following meals with differing macronutrient compositions. As IR2's first deployment, the device lacked the IR amplifier system, collecting 228 hours of spectral data across 19 subjects at the UVA Diet and Nutrition lab. Tab. II shows the device configuration.

Although the nutritional findings were inconclusive, this deployment informed multiple improvements to subsequent iterations of the device. The current generated by the IR photodiode was too small for the ADC on the nRF52840, resulting in IR measurements dominated by ADC noise rather than meaningful spectral information. In the next iteration, we added a transimpedance amplifier to the IR photodiode to increase the signal amplitude before ADC acquisition. Deployment also revealed several challenges: variable fit across wrist sizes, accidental impacts against hard surfaces, limited signal range restricting participant mobility, and water exposure during handwashing. On the operational side, managing participant ID tracking with a small device pool and ensuring devices were recharged between sessions proved difficult.

V. CONCLUSION AND FUTURE WORK

While IR2 has been deployed in multiple lab and clinical environments, its range of use is still limited. First, the current sensing footprint is too large to support a fingertip form factor commonly used in clinical applications such as pulse oximetry. Second, the device's height limits wearability due to battery thickness, even though its width and length match typical consumer smartwatches. Third, the housing for IR2 is not waterproof, limiting the environments the device can be deployed in. Fourth, manufacturing or calibration improvements are needed to reduce inter-device variation. Finally,

future iterations may improve device granularity by expanding the ADC from 12-bits and by exploring differences between analog versus PWM dimming methods for the LED driver.

In this paper, we presented IR2 a visible to SWIR wearable spectroscopy device. We ran four experiments to verify device performance: a device-to-device comparison, a battery life test across all operational configurations, a temperature experiment, and a fluid study. We then deployed IR2 across four lab-controlled and clinical studies, collecting over 560 hours of optical data on 79 participants across exercise, clinical, nutrition, and eating applications.

VI. ACKNOWLEDGMENT

The authors thank V. Bruot, A. Gay, V. Pierce, B. Roberts, and J. Wallemacq for their contributions to the Nutrition study. This research was supported by the University of Virginia's Wallace H. Coulter Center for Translational Research and the University of Virginia Engineering Research Innovation Award.

REFERENCES

- [1] F. Kurul, D. Aydoğan, S. Janat, I. Kirlangic, H. Kaya, and S. Topkaya, "Wearable sensors for health monitoring: Current applications, trends, and future directions," *Biosensors and Bioelectronics*: X, 2025.
- [2] A. Smith, R. Li, and Z. Tse, "Reshaping healthcare with wearable biosensors," *Scientific Reports*, p. 4998, 2023.
- [3] E. Polat, M. Cetin, A. Tabak, E. Bilget Güven, B. Uysal, T. Arsan, A. Kabbani, H. Hamed, and S. Gül, "Transducer technologies for biosensors and their wearable applications," *Biosensors*, 2022.
- [4] K. Budidha, M. Mamouei, N. Baishya, M. Qassem, P. Vadgama, and P. Kyriacou, "Identification and quantitative determination of lactate using optical spectroscopy—towards a noninvasive tool for early recognition of sepsis," *Sensors*, 2020.
- [5] H. Zhang, E. Courtney, K. C. Quinn, and A. Watson, "Openspectro: An open-source spectroscopic profiling platform," in *2025 47th Annual International Conference of the IEEE Engineering in Medicine and Biology Society*. IEEE, 2025.
- [6] S. Spink, A. Pilvar, L. Wei, J. Frias, K. Anders, S. Franco, O. C. Rose, M. Freeman, G. Bag, H. Huang *et al.*, "Shortwave infrared diffuse optical wearable probe for quantification of water and lipid content in emulsion phantoms using deep learning," *Journal of Biomedical Optics*, 2023.
- [7] A. Watson, C. Kendell, A. Lingamoorthy, I. Lee, and J. Weimer, "Lumos: An open-source device for wearable spectroscopy research," *Proceedings of the ACM on Interactive, Mobile, Wearable and Ubiquitous Technologies*, 2023.
- [8] L. Lederer, A. Roghanizad, R. Janamsetty, L. Redmore, S. Kim, K. Bansal, A. Liu, J. Dunn *et al.*, "Vitalwave: An end-to-end open-source high-frequency wearable device and data collection platform," in *21st International Conference on Body Sensor Networks*. IEEE, 2025.
- [9] T. Hamid, P. Flores, J. Byun, E. Courtney, K. Quinn, and A. Watson, "Dermaglow: Objective quantification of melanin, erythema and skin-tone using wearable optical spectroscopy," *Proceedings of the ACM on Interactive, Mobile, Wearable and Ubiquitous Technologies*, 2025.
- [10] T. Hamid, P. Flores, J. Byun, X. Chen, H. Zhang, K. Quinn, and A. Watson, "A multi-wavelength optical sensing framework for calibration-free wearable blood pressure monitoring," in *IEEE International Conference on Acoustics, Speech and Signal Processing (ICASSP)*, 2025.
- [11] T. Hamid, E. C. Courtney, P. Flores, J. Byun, A. Doryab, S. Kranz, and A. Watson, "Characterization and feasibility of wearable spectroscopic tracking of nutrition biomarkers," *IEEE Pervasive Computing*, 2025.
- [12] V. Rybnyok, J. May, K. Budidha, and P. Kyriacou, "Design and development of a novel multi-channel photoplethysmographic research system," in *Point-of-Care Healthcare Technologies*. IEEE, 2013.
- [13] J. D'orazio, S. Jarrett, A. Amaro-Ortiz, and T. Scott, "Uv radiation and the skin," *International journal of molecular sciences*, 2013.
- [14] A. Douplik, G. Saiko, I. Schelkanova, and V. Tuchin, "The response of tissue to laser light," in *Lasers for medical applications*. Elsevier, 2013.